

71261 - Seal Endurance Test with Oil Fill

Endurance test for Pinion Application – VECV

Anoop N

Global Testing India - Bangalore

TR25-0009-BTS

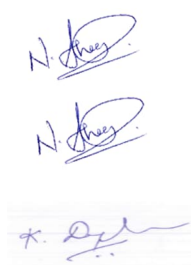
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1 Objective

To conduct the Endurance Test with Oil fill on 3 samples of 71261 seals to check the performance of the seal for the application pinion oil seal – VECV.

2 Conclusions

Sl. No	Sample #	Target Hours	Completed Hours	Result	Remarks
1	71261 – S#01	960	960	PASS	Zero Leakage observed
2	71261 – S#03	960	960	PASS	Zero Leakage observed
3	71261 – S#04	960	960	PASS	Zero Leakage observed

3 Test samples identification

3.1 Date of receipt of the samples

- 03 Samples each of 71261 were received on 21.11.2024.

3.2 Identification of the samples



71261

Sample No.	Manufacturing location
71261 S#01, S#02, S#03	SKF Sealing Solutions, Mysore

3.3 Description and condition of the test samples

- Samples were wrapped with packing materials and found in good conditions (without any damage).

4 Test description

4.1 Description/specification of the test

Sample Designation.	Specification as per OSTR/Drawing	71261 – Sample No. 01	71261 – Sample No. 03	71261 – Sample No. 04
Test Request Form #	Test Request form 25-01-71261	Test Request form 25-01-71261		
Total Duration [Hrs.]	960	960	960	960
Speed [rpm]	Refer Test cycle (after this table)	Refer Test cycle		
Direction of Rotation	Refer Test cycle (after this table)	Refer Test cycle		
Dynamic Run out [mm]	0.10 ± 0.025 mm	0.09	0.09	0.09
Static Run out [mm]	0.10 ± 0.025 mm	0.09	0.09	0.09
Seal Cock [mm]	0.15 ± 0.025 mm	0.13	0.14	0.14
Shaft Surface Finish (µm Ra)	0.20 ~ 0.50 µm Ra	0.40~0.45	0.41~0.42	0.253~0.286
Shaft Diameter (mm)	55.9630 ± 0.03mm	55.97	55.96	55.95
Shaft hardness (HRC)	35 HRC Min.	45.2	46.7	40.2
Housing Diameter (mm)	99.0175 ± 0.01 mm	99.01	99.03	99.02
Housing Surface Finish (µm Ra)	1.6 ~ 2.5 µm Ra	1.50~1.52	1.53~1.75	1.66~1.68
Fluid	Servo Gear Super 85W 140	Servo Gear Super 85W 140		
Temperature (°C)	Refer Test cycle (after this table)	Refer Test cycle (after this table)		
Fluid level	Shaft Centreline	Shaft Centreline		
Oil change interval	240 Hours	240 Hrs	240 Hrs	240 Hrs

Test Procedure:

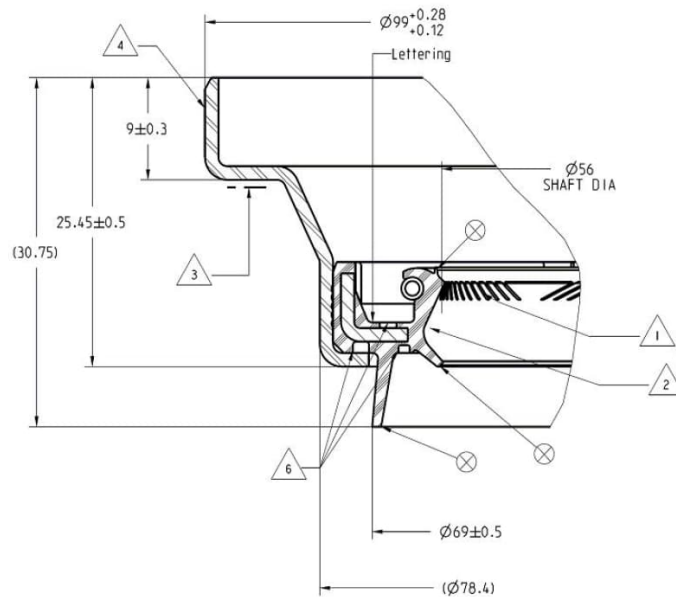
- Pre and Post-test measurements of Seals are as follows,

Measurement		Pre	Post	Measurement		Pre	Post
M1	Primary Seal Lip Diameter (With Spring)	YES	YES	M14		N/A	N/A
M2	Primary Seal Lip Diameter (No Spring)	N/A	N/A	M15		N/A	N/A
M3	Radial Load (With Spring)	YES	YES	M16		N/A	N/A
M4	Radial Load (No Spring)	N/A	N/A	M17		N/A	N/A
M5	Radial Dust Lip Diameter	YES	YES	M18		N/A	N/A
M6	Rubber Outside Diameter	N/A	N/A	M19		N/A	N/A
M7	Metal Outside Diameter	YES	YES	M20		N/A	N/A
M8	Axial Lip Diameter	YES	YES	M21		N/A	N/A
M9	Axial Lip Height	YES	YES	Q1	Photographs (Describe Below)	YES	YES
M10	Static Band Wear	N/A	YES	Q2	Shaft Wear Data	N/A	YES
M11	Primary Lip Wear Track	N/A	YES	Q3	Return Shaft With Seal	N/A	N/A
M12	Radial Excluder Lip Wear	N/A	YES	Q4		N/A	N/A
M13	Axial Lip Wear	N/A	N/A	Q5		N/A	N/A

- Tolerance for Speed is ± 50 RPM and Temperature is $\pm 5^{\circ}\text{C}$
- Test cycle Will be as follows:

Activity or Operation	Segment Number	Segment Time			Direction CW, CCW, FWD, REV, Apply, Return, In, Out	Speed Start of Segment Or N/A	Speed End of Segment	Temperature Start of Segment Or N/A	Temperature End of Segment	Pressure Start of Segment Or N/A	Pressure End of Segment	Contamination	Reciprocation	Lubrication / Height from Spindle Half Shaft (ex. +/- 0.25 inch)
		HH	MM	SS										
O	1				N/A									
O	2	19	0	0	CW	0	3800	Ambient	70	Ambient	Ambient	NA	NA	Half
O	3	0	30	0	CW	3800	4500	70	100	Ambient	Ambient	NA	NA	Half
O	4	0	30	0	CCW	4500	800	100	70	Ambient	Ambient	NA	NA	Half
O	5	4	0	0	N/A	800	0	70	Ambient	Ambient	Ambient	NA	NA	Half
A	6	Repeat above cycle 40 Times (960 Hours)												

- Total test duration for each sample is 960 hours or until failure (whichever is earlier).
- Test setup and seal photographs shall be shared in the test report.



Acceptance Criteria:

- No Appearance of T4 Oil leakage throughout the test.

Procedure for monitoring:

- Each day in the morning between 8 a.m. and 9 a.m. to examine the seal (At the interference between the shaft OD and the seal ID, at interference between the bore ID and the seal OD) area for the oil seepage or leakage which constitutes a failure.
- On Friday of every week, the observation will be made again between 3.30 pm to 4.30 pm.
- During the weekend (Saturday & Sunday) and public holidays, the visual inspection cannot be done, and any leakage found on the next inspection day, the total hours accumulated will be considered as the previous inspected completed hours.

Disclaimer:

Seals dimensional results will be reported as actual, and no inference will be made on the results.

4.2 Test and measuring equipment.



Endurance Testing Machine

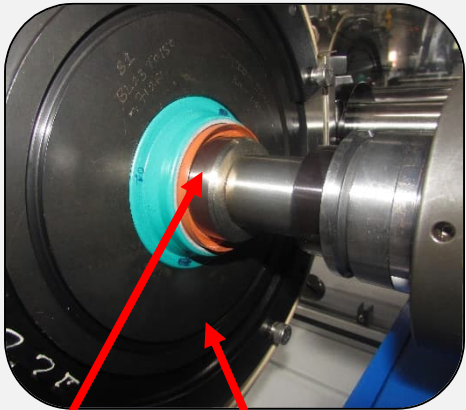
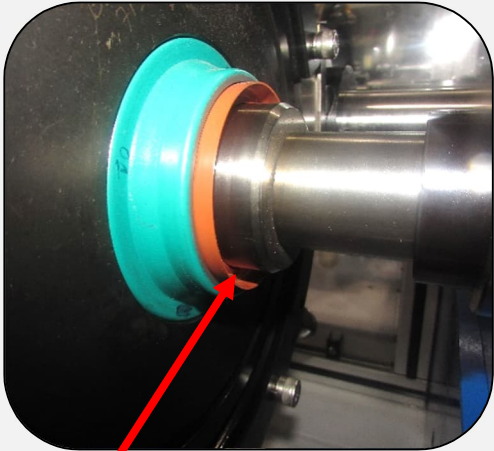
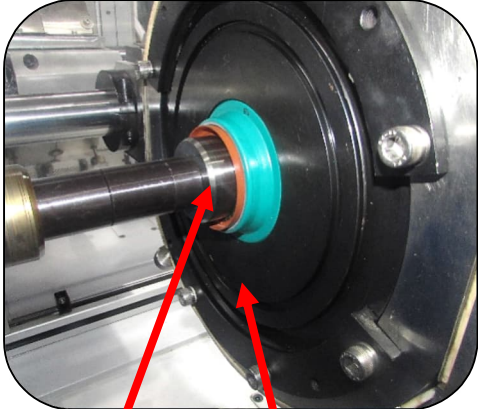
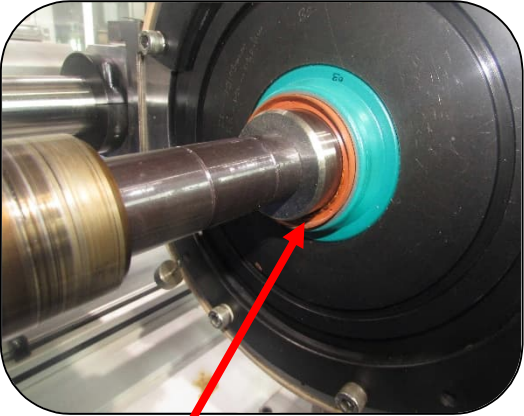
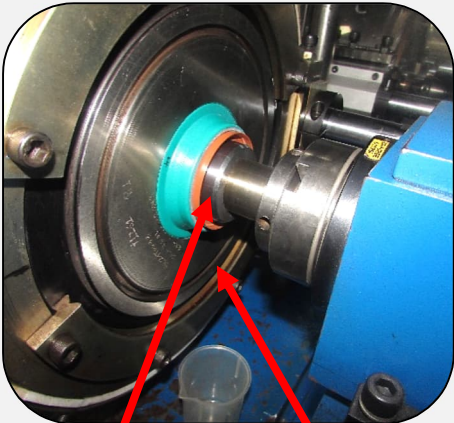
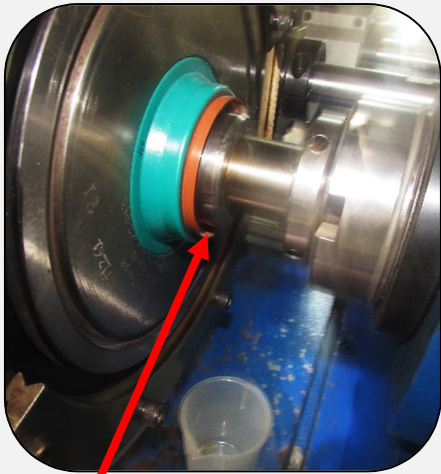
4.3 Date the test was performed.

Sample Number	Start Date	End Date
71261 - S#01	15 th January 2025	24 th February 2025
71261 - S#02	4 th February 2025	17 th March 2025
71261 - S#03	17 th February 2025	31 st March 2025

5 Test results

Important: Unless otherwise indicated - the results presented in this report relate only to the items that are tested.

3 samples were run as per the test cycle (Refer the test specification) and **all the sample found with zero Leakage at the main lip**. Hence, all the samples are considered to have **PASS** the test.

Sample #	Photos – Before Test	Photos – After Test
71261 S#01	 Shaft Bore	 Zero leakage observed
71261 S#03	 Shaft Bore	 Zero leakage observed
71261 S#04	 Shaft Bore	 Zero leakage observed

Pre and Post Test measurement results:

Sl. No	Description	Spec (as per Drwg. / OSTR)	Measuring Equipment	Pre-Test Measurement - 71261		
				S#01	S#03	S#04
1	Main lip Dia (Sprung) (mm)	NA	Non-Contact type	54.050	54.024	54.028
2	Metal Outer Dia (mm)	NA	Non-Contact type	99.192	99.187	99.192
3	Dirt lip Dia (Felt Lip)- mm	NA	Non-Contact type	56.289	56.330	56.276
4	Radial load (Sprung) – N/cm	NA	Radial Load Instrument	1.330	1.377	1.351

Sl. No	Description	Spec (as per Drwg. / OSTR)	Measuring Equipment	Post-Test Measurement - 71261		
				S#01	S#03	S#04
1	Main lip Dia (Sprung) (mm)	NA	Non-Contact type	54.143	54.079	54.084
2	Metal Outer Dia (mm)	NA	Non-Contact type	99.164	99.137	99.135
3	Dirt lip Dia (Felt Lip)- mm	NA	Non-Contact type	56.624	56.555	56.094
4	Radial load (Sprung) – N/cm	NA	Radial Load Instrument	1.252	1.079	1.195